

Aman Malik

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Objective

Ambitious Computer Science undergraduate with hands-on experience in computer vision, deep learning, and full-stack development. Passionate about Artificial Intelligence and Machine Learning, with practical experience building CNN-based classification systems, object detection pipelines, and retrieval-augmented generation (RAG) applications. Seeking an AI-focused role where I can contribute to building intelligent systems and agentic AI applications while solving real-world challenges.

Education

08/2023 – Present
Karachi, Pakistan

Bachelor of Science in Computer Science

FAST NUCES
CGPA: 3.73

08/2021 – 05/2023
Karachi, Pakistan

A levels

Highbrow School & College

- Mathematic: A
- Physics: A
- Chemistry: B

Technical Skills

Languages

Python, C/C++, SQL, PHP, PostgreSQL

Systems & Tools

Linux/ Bash, Git, POSIX Threads, Concurrency, InterProcess Communication

Frameworks & Tools: PyTorch, TensorFlow/Keras, OpenCV, YOLOv8, LangChain, NumPy, Matplotlib, Scikit-Learn, Streamlit

AI & Mathematics

CNNs, RAG (Retrieval-Augmented Generation), Vector Databases (ChromaDB), Neural Collaborative Filtering, Matrix Factorization, Cold-Start Mitigation, Linear Algebra, Multivariable Calculus, Probability & Statistics

Projects

AI-Powered Multilingual Translator

Python | Hugging Face Transformers | Meta NLLB-200 | Streamlit

- Developed an AI-powered multilingual translation application using Meta NLLB-200, enabling translation across 200+ languages with an interactive Streamlit interface.
- Implemented transformer-based neural machine translation with Hugging Face Transformers and optimized inference using model caching for efficient performance.

Face Emotion Detection System

CNN, TensorFlow/Keras, OpenCV

- Designed and trained a custom CNN from scratch on the FER2013 dataset (35K+ images) to classify facial expressions into 7 emotion categories, achieving 67.6% test accuracy using data augmentation, batch normalization, and dropout regularization
- Built a real-time emotion detection pipeline using OpenCV for face detection, integrated with the trained model to support both live webcam and static image inference

Object Detection System(YOLO)

Deep Learning

- Built an object detection system using YOLOv8 architecture with transfer learning on a custom MS COCO dataset
- Trained and evaluated the model on curated datasets, analyzing detection performance across multiple object categories
- Applied computer vision techniques including annotation preprocessing and bounding box conversion for real-world detection scenarios

Awards & Achievements

Dean's List Honors

Outstanding academic performance in 1st, 2nd, 3rd, 4th,5th and 6th semesters

Programming Competitions

Participated in DevDays and Coder's Cup, showing problem-solving and teamwork